Process Schedulers

- Schedulers are special system software which handle process scheduling in various ways.

1. **Long-term scheduler** (or **job scheduler**) –

- It mainly moves processes from Job Queue to Ready Queue.

- It controls the Degree of Multi-programming, i.e., the number of processes present in a ready state or in main memory at any point in time.

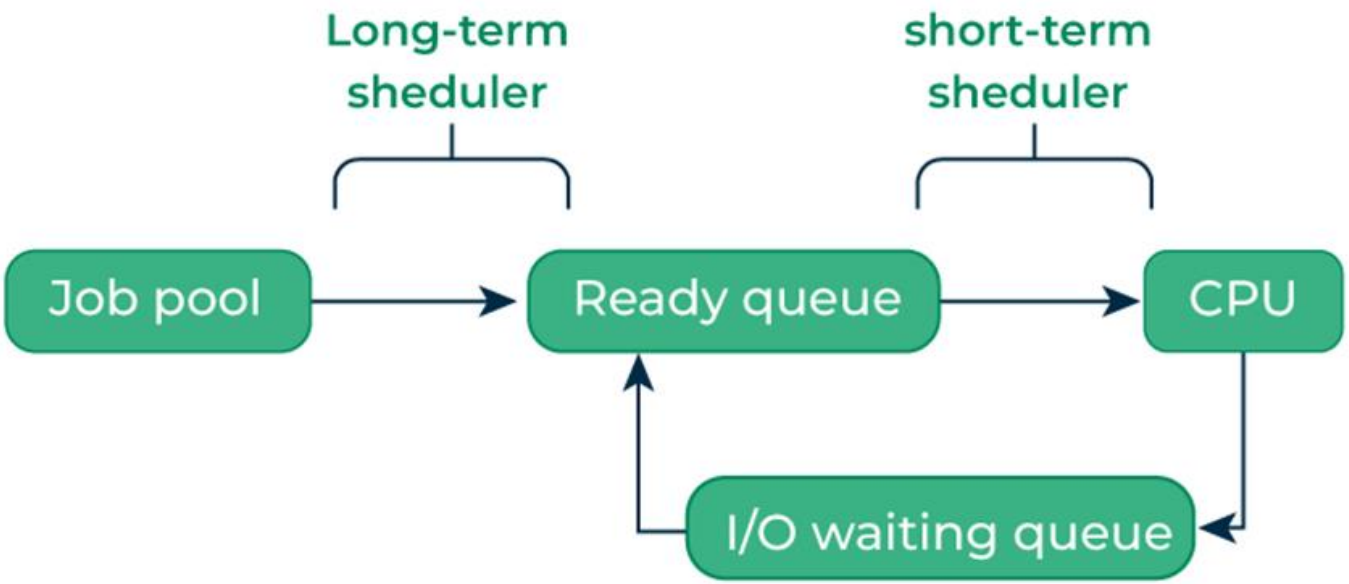
- It is important that the long-term scheduler make a careful selection of both I/O and CPU-bound processes. I/O-bound tasks are which use much of their time in input and output operations while CPU-bound processes are which spend their time on the CPU. The job scheduler increases efficiency by maintaining a balance between the two.

- In some systems, the long-term scheduler might not even exist. For example, in time-sharing systems like Microsoft Windows, there is usually no long-term scheduler. Instead, every new process is directly added to memory for the short-term scheduler to handle.

- Slowest among the three (that is why called long term).

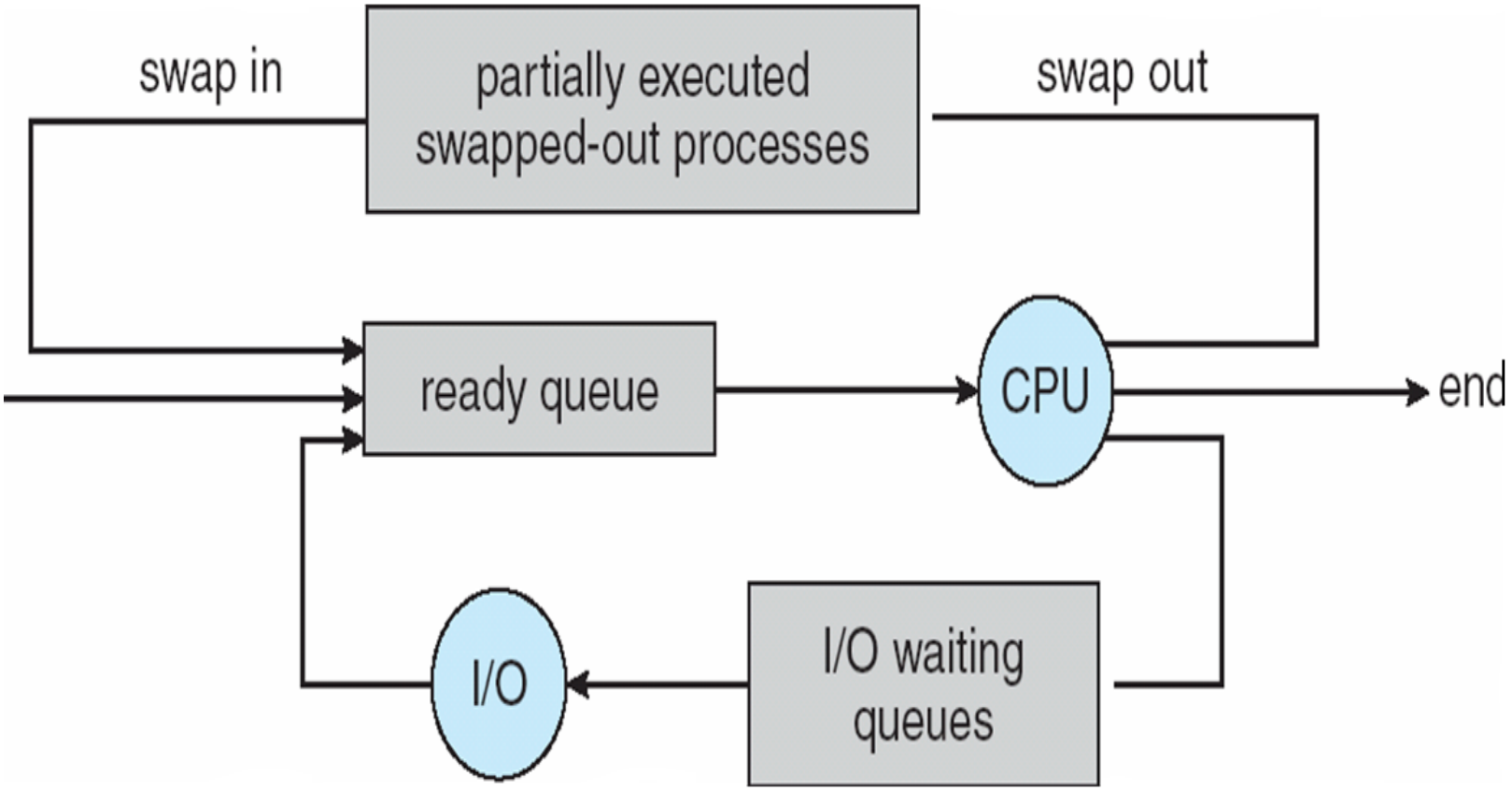
2.Short-term scheduler (or **CPU scheduler**) –CPU Scheduler is responsible for selecting one process from the ready state for running (or assigning CPU to it).

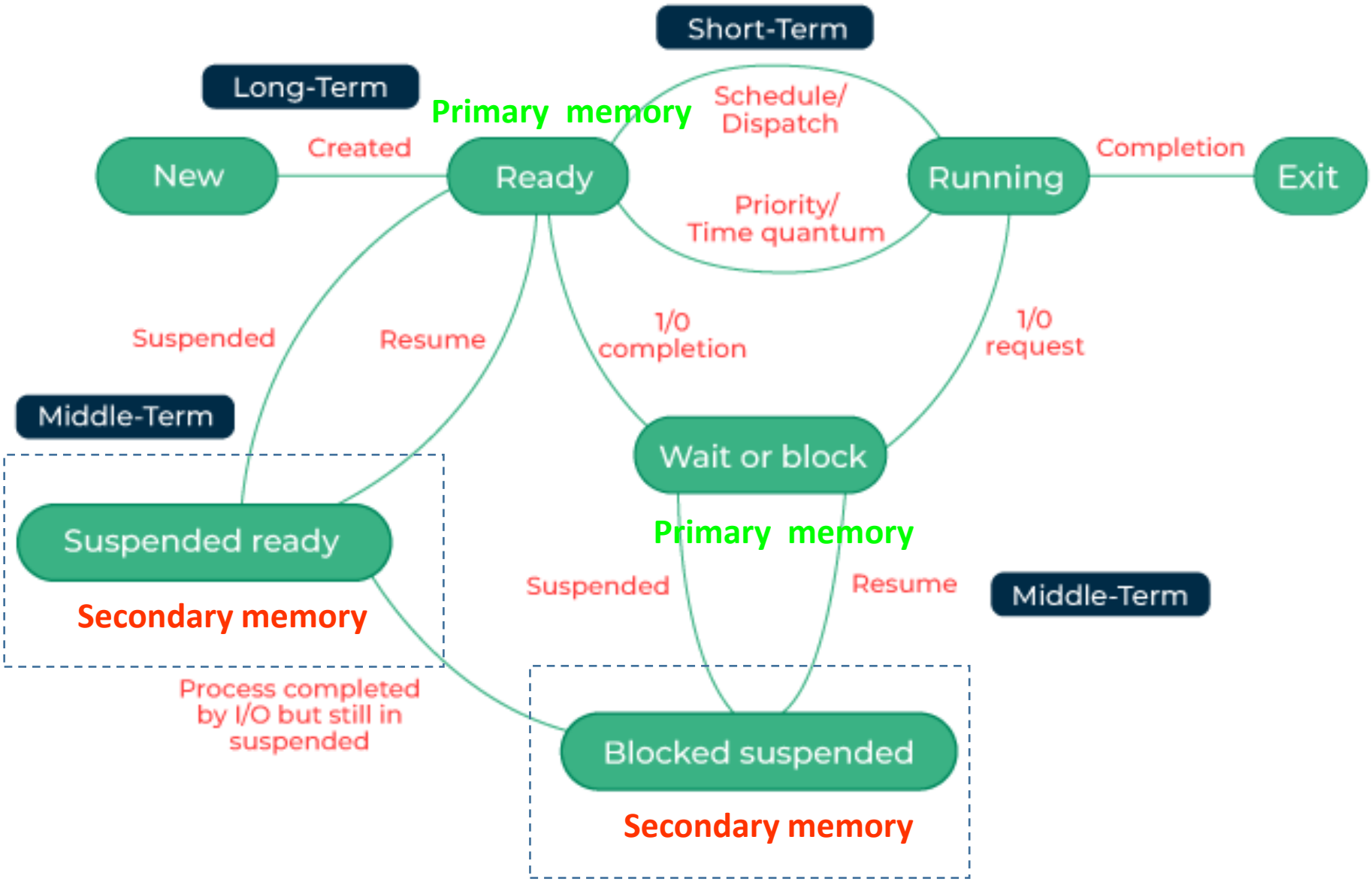
- STS (Short Term Scheduler) must select a new process for the CPU frequently to avoid starvation(occurs when a process in the OS runs out of resources because other processes are using it).
- The CPU scheduler uses different scheduling algorithms to balance the allocation of CPU time, like- FCFS, SJF, Round Robin.
- It picks a process from ready queue.
- Its main objective is to make the best use of CPU.
- It mainly calls dispatcher.
- Fastest among the three (that is why called Short Term).


Long-term
sheduler
short-term
sheduler
Job pool
Ready queue
CPU
I/O waiting queue

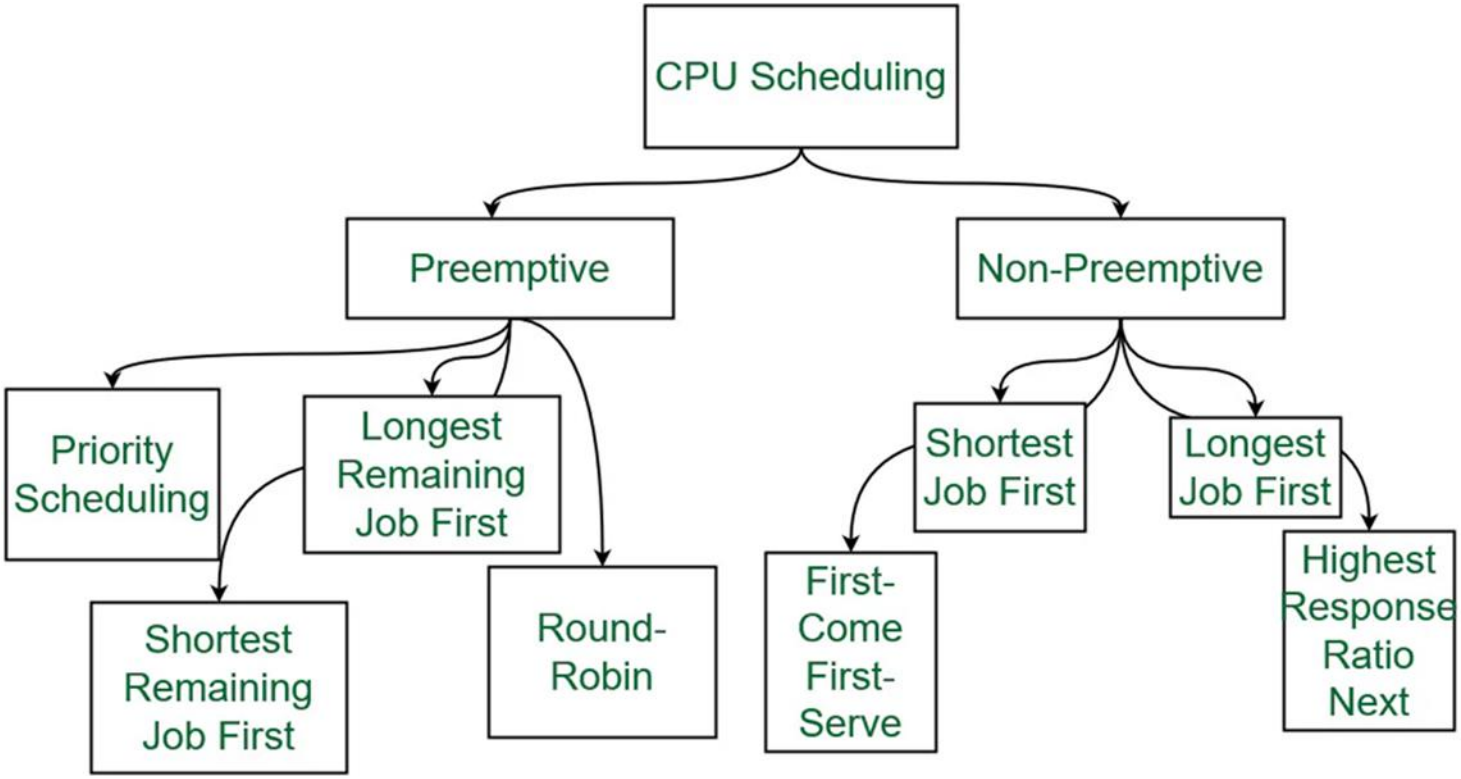
3.Medium-term scheduler -Medium Term Scheduler (MTS) is responsible for moving a process from memory to disk (or swapping).

- It reduces the degree of multiprogramming (Number of processes present in main memory).
- A running process may become suspended if it makes an I/O request. A suspended processes cannot make any progress towards completion. In this condition, to remove the process from memory and make space for other processes, the suspended process is moved to the secondary storage. This process is called swapping, and the process is said to be swapped out or rolled out. Swapping may be necessary to improve the process mix (of CPU bound and IO bound)
- When needed, it brings process back into memory and pick up right where it left off.
- It is faster than long term and slower than short term


swap in
partially executed
swapped-out processes
swap out
ready queue
CPU
end
I/O
I/O waiting
queues


Short-Term
Long-Term
Primary memory
Schedule/
Dispatch
New
Created
Ready
Running
Completion
Exit
Priority/
Time quantum
Suspended
Resume
I/O
completion
I/O
request
Middle-Term
Wait or block
Suspended ready
Primary memory
Suspended
Resume
Middle-Term
Secondary memory
Process completed
by I/O but still in
suspended
Blocked suspended
Secondary memory

- A Process Scheduler schedules different processes to be assigned to the CPU based on particular scheduling algorithms. These algorithms are either **non-preemptive or preemptive.**
- **Preemptive Scheduling:** a CPU scheduling method where a running process can be interrupted (preempted) by another process with a higher priority or one that needs urgent attention. Preemptive scheduling is used when a process switches from running state to ready state or from the waiting state to the ready state.
- **Non-Preemptive Scheduling:** on the other hand, once a process is given access to the CPU, it cannot be interrupted or preempted until it either completes its execution or voluntarily relinquishes the CPU by entering a waiting state. Non-Preemptive scheduling is used when a process terminates , or when a process switches from running state to waiting state.


CPU Scheduling
Preemptive
Non-Preemptive
Priority Scheduling
Longest Remaining Job First
Shortest Remaining Job First
Round-Robin
Shortest Job First
Longest Job First
First-Come First-Serve
Highest Response Ratio Next

Parameter	PREEMPTIVE SCHEDULING	NON-PREEMPTIVE SCHEDULING
Basic	In this resources(CPU Cycle) are allocated to a process for a limited time.	Once resources(CPU Cycle) are allocated to a process, the process holds it till it completes its burst time or switches to waiting state
Interrupt	Process can be interrupted in between.	Process can not be interrupted until it terminates itself or its time is up
Starvation	If a process having high priority frequently arrives in the ready queue, a low priority process may starve	If a process with a long burst time is running CPU, then later coming process with less CPU burst time may starve
Flexibility	flexible	Rigid

Cost	Cost associated	No cost associated
CPU Utilization	In preemptive scheduling, CPU utilization is high	It is low in non preemptive scheduling
Waiting Time	Preemptive scheduling waiting time is less	Non-preemptive scheduling waiting time is high
Response Time	Preemptive scheduling response time is less	Non-preemptive scheduling response time is high
Decision making	Decisions are made by the scheduler and are based on priority and time slice allocation	Decisions are made by the process itself and the OS just follows the process's instructions
Process control	The OS has greater control over the scheduling of processes	The OS has less control over the scheduling of processes
Overhead	Higher overhead due to frequent context switching	Lower overhead since context switching is less frequent

Different time Concepts in CPU Scheduling

- **Arrival Time:**

In CPU Scheduling, **the arrival time refers to the moment in time when a process enters the ready queue and is awaiting execution by the CPU**.

- **Burst Time:**

Burst time, also referred to as "execution time". **It is the amount of CPU time the process requires to complete its execution.**

It's a duration of time.

- **Completion Time:**

Completion time is when a process finishes execution and is no longer being processed by the CPU. It is the summation of the arrival, waiting, and burst times.

- **Turnaround Time:**

The time elapsed between the arrival of a process and its completion is known as turnaround time. That is, the duration it takes for a process to complete its execution and leave the system.

Turnaround Time = Completion Time – Arrival Time

- **Waiting Time:**

This is a process's duration in the ready queue before it begins executing. It helps assess how efficient the scheduling algorithm is. A scheduling method that consistently results in reduced wait times for processes, for example, is considered more efficient than one that regularly results in longer wait times.

Waiting Time = Turnaround Time – BurstTime

Furthermore, it helps to measure the efficiency of a scheduling algorithm.

- **Response Time:**

Response time is the amount of time it takes for the CPU to respond to a request made by a process. **It is the duration between the arrival of a process and the first time it runs.** It is an essential parameter in CPU scheduling since it may assist in determining a system's perceived responsiveness to user requests.

Response Time = Time it Started Executing – Arrival Time